

Gopalakrishna Maddipalli

GENERATIVE AI ENGINEER | DATA SCIENTIST

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PROFILE

Production-minded AI engineer with seven-plus years spanning construction operations, research, data science, and applied AI. Builds evidence-grounded RAG systems, multi-agent workflows, document intelligence, predictive models, and cloud APIs with explicit evaluation and human review.

CORE CAPABILITIES

Generative AI	LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, MCP, prompt engineering
Retrieval	Qdrant, Pinecone, Milvus, Chroma, FAISS, hybrid search, reranking, evaluation
Models & ML	Llama, Qwen, DeepSeek, PyTorch, scikit-learn, XGBoost, SHAP, LoRA / QLoRA
Engineering	Python, FastAPI, Docker, AWS, Azure Container Apps, Next.js, TypeScript, GitHub Actions
Data & documents	OCR, Textract, PaddleOCR, spaCy, PostgreSQL, PySpark, Athena, Power BI

EXPERIENCE

AI/ML Engineer

Sep 2024 - Present

SunRay Construction Solutions | Hyderabad, Telangana, India

- Build and implement AI-powered applications for construction legal operations, research, and compliance workflows.
- Develop evidence-aware retrieval, agentic, document-intelligence, and predictive workflows with human review.
- Connect operational evidence, model outputs, evaluation, and cloud APIs through production-minded engineering.

Data Scientist

2021 - 2024

SunRay Construction Solutions | Hyderabad, Telangana, India

- Applied data science and machine learning to construction operations, research, and payment-protection workflows.
- Translated operational data into analytical and predictive workflows following postgraduate study in AI and machine learning.

Research Analyst

2019 - 2021

SunRay Construction Solutions | Hyderabad, Telangana, India

- Supported construction legal research and operational work-order processing.
- Emphasized accurate source verification and documented decision-making across research workflows.

Junior CAD Engineer

May 2019 - Sep 2019

Venusgeo Solutions | India

- Produced accurate building and school-facility information for a UK county-council asset-management project.

SELECTED AI SYSTEMS

Autonomous Work Order Intelligence & Operations Platform

Microsoft Foundry, Python, FastAPI, MCP, AWS, Azure Container Apps

- Evidence-aware multi-agent architecture for intake, research, discrepancy detection, controlled correction, and human escalation.

NTO Operations Copilot

Microsoft Foundry, MCP, FastAPI, Next.js, AWS Lambda, Azure App Service

- Evidence-grounded research coach guiding new users through notice selection, exceptions, and approved next actions.

Construction Payment Risk Prediction

Python, FastAPI, AWS, DataRobot, SHAP

- Human-reviewed operational prioritization with explainable risk signals for payment-protection workflows.

Construction Document Intelligence

Textract, PaddleOCR, spaCy, PostgreSQL, FastAPI

- Structured, reviewable extraction from complex legal PDFs to reduce manual document processing.

EDUCATION & SELECTED CREDENTIALS

Post Graduate Program in Artificial Intelligence and Machine Learning

McCombs School of Business, The University of Texas at Austin via Great Learning | Jan 2021

Bachelor of Technology - Mechanical Engineering

YSR Engineering College of YVU | 2014 - 2018

Agent Architect - Founderz AI & Business School | Sep 2026

Generative AI for Data Scientists - IBM / Coursera Specialization | Sep 2026

Generative AI for Data Analysts - IBM / Coursera Specialization | Sep 2026

Generative AI: Elevate your Software Development Career - IBM / Coursera | Sep 2026

Machine Learning Engineering for Production (MLOps) - Coursera Specialization | Nov 2021

Practical Data Science - Coursera Specialization | Sep 2021

Neural Networks and Deep Learning - Coursera | Sep 2020